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Sex and boldness explain individual differences in spatial learning in a lizard

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Understanding individual differences in cognitive performance is a major challenge to animal behaviour and cognition studies. We used the Eastern water skink (*Eulamprus quoyii*) to examine associations between exploration, boldness and individual variability in spatial learning, a dimension of lizard cognition with important bearing on fitness. We show that males perform better than females in a biologically relevant spatial learning task. This is the first evidence for sex differences in learning in a reptile, and we argue that it is probably owing to sex-specific selective pressures that may be widespread in lizards. Across the sexes, we found a clear association between boldness after a simulated predatory attack and the probability of learning the spatial task. In contrast to previous studies, we found a nonlinear association between boldness and learning: both 'bold' and 'shy' behavioural types were more successful learners than intermediate males. Our results do not fit with recent predictions suggesting that individual differences in learning may be linked with behavioural types via high–low-risk/reward trade-offs. We suggest the possibility that differences in spatial cognitive performance may arise in lizards as a consequence of the distinct environmental variability and complexity experienced by individuals as a result of their sex and social tactics.

1. Introduction

A fundamental aim in cognitive studies is to understand the factors that might explain the extraordinary levels of individual variability in cognitive performance observed in almost every animal species thus far studied [1–3]. Recent research has made some progress in this respect, but we are only just beginning to understand how cognitive performance relates to development and selection at the intraspecific level [4,5]. In sharp contrast, the study of intraspecific variation in non-cognitive-behavioural traits is a thriving area of research. The study of behavioural types or personality traits (i.e. consistent behavioural tendencies across time and context) has driven our understanding of intraspecific behavioural differences during the past decade, generating several hypotheses about the evolution of adaptive behavioural variation at the individual level [6–11]. Interestingly, recent hypotheses have proposed that learning and non-cognitive-behavioural traits may covary as part of the same suite of correlated traits. Furthermore, they can both determine the environment that is to be experienced by different individuals, generating feedback loops that may equally lead to cognitive–behavioural syndromes [4,5,8,9,12–15].

Sih & Del Giudice [4] recently proposed that variation in cognition and personality may be functionally related through the existence of a shared risk–reward trade-off between fast–slow behavioural traits and speed–accuracy cognitive styles [4]. Many behavioural traits can be classified into a fast–slow axis (e.g. bold versus shy, proactive versus reactive, fast versus slow exploration), and this variation can be associated with variation in a risk–reward axis because bolder, more aggressive, exploratory and/or proactive individuals have a greater potential to gather resources, but take more risks in doing so [4,15–17]. Similarly, speed–accuracy trade-offs are bound to affect cognitive styles, because animals

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that learn fast do so at the expense of acquiring inaccurate information [18]. The speed–accuracy trade-off is also essentially related to variation in the risk–reward axis because fast learning is inherently risky (i.e. based on inaccurate information) but will tend to draw more resources in the short term [4]. The overarching idea of the risk–reward hypothesis is that selection for factors leading to the adoption of a more risk-prone lifestyle will result in correlated selection for both faster behavioural traits and faster but less accurate and flexible learning, and *vice versa* [19]. For example, risk-prone individuals may be selected for in stable local habitats, where fast exploration would give them a competitive advantage and the evolution and/or development of learning abilities would aid in the quick formation of routines, whereas risk-averse individuals with learning abilities that are more flexible and sensitive to environmental change may develop or be selected for in more variable local habitats [20]. This hypothesis has found some support in a few bird and fish species, in which proactive individuals tend to be quicker than reactive individuals at operant learning tasks and in avoidance learning tasks fundamentally guided by external environmental cues [4,21–23], but slower in reversal learning tasks [20,24]. In short, recent advances suggest that an obvious avenue for understanding adaptive individual variation in learning is to study the existence of covariation between cognition and non-cognitive–behavioural traits, ideally in an ecologically relevant context where learning may directly impinge on individual fitness.

Sex is an equally important factor in understanding individual variation in learning and in associated behavioural types. The sexes will frequently experience different environmental complexity and/or variability as a consequence of their different reproductive strategies, which are likely to drive differences in both cognitive and non-cognitive traits, and in the way they covary [24–26]. However, while sex differences in learning have been well documented in some taxa [3], they have been completely neglected in others, such as lizards (and reptiles at large). Even less information is available about sex differences in personality traits, which have been documented in only a handful of fish and bird species [26–29]. Finally, scarcely any attention at all has been paid to studying sexual differences in the existence and form of cognitive–behavioural types [24,30], despite the fact that there are sound theoretical reasons to expect them. In the context of the risk–reward hypothesis, for example, males may frequently be forced to adopt more risky reproductive strategies than females because of their different sexual roles, and this could lead to general sex differences in learning, personality traits and their covariation [31].

Spatial learning is a cognitive dimension likely to be of utmost importance to lizards [32]. It is believed to be under strong selection given its importance in foraging, territorial and anti-predatory behaviour, which often require quick and flexible learning of territorial boundaries, suitable escape routes and refuges [33,34]. Not surprisingly, lizards have been found to be capable of quick and flexible spatial learning when tested under a biologically realistic learning paradigm [34]. Furthermore, males and females of many lizard species are generally subjected to different spatial demands because of differences in reproductive tactics and behaviour during the reproductive season. These sex-specific tactics and behaviours may have given rise to widespread sexual differences in spatial learning abilities, and to sex-specific associations between spatial learning and other behavioural traits [24,25]. An additional dimension in many lizard systems is that exploratory and boldness traits may covary with alternative reproductive tactics (hereafter ARTs) [35]. Because lizard ARTs are closely associated with territorial behaviour, these traits represent ecologically significant behavioural variation in a context in which spatial learning is important for lizards [36].

Here, we used an Australian lizard, the Eastern Water Skink (*Eulamprus quoyii*), to explore associations between individual variability in spatial learning performance, sex and exploratory and boldness traits that have been previously identified as important covariates of ARTs in *E. quoyii* and *Eulamprus heatwolei*, a closely related species [37–41]. Our main aims in this study were: (i) to examine the existence of sexual differences in learning performance and (ii) to explore the existence, form and potential sex differences in associations between spatial learning, exploration and boldness. In order to do so, we assayed behavioural and cognitive traits in four successive experiments in which we quantified exploratory behaviour, boldness in two different contexts (neophobia towards novel prey and boldness after a predatory attack) and performance in a simple spatial task.

2. Material and methods

(a) Study species

The Eastern water skink (*E. quoyii*) is a large (90–122 mm snout–vent length (SVL)), viviparous lizard species that is widely distributed across Eastern Australia, from South Australia and Victoria through to New South Wales and into Queensland. It frequently inhabits rocky water edges in suburban areas and can reach high densities. We collected 216 water skinks in August and September 2010 from five separate sites throughout the Sydney region as part of a separate natural mating experiment that took place during the breeding season (see [42] for details). After the breeding season, all lizards were transferred to large outdoor bins (3.2 m diameter) containing bark mulch substrate, logs and roofing tiles for refuges. Lizards had constant access to water and were fed crickets every second day.

We used 64 of these lizards (32 males and 32 females) in March 2011 in our experiments. Experiments commenced immediately after transferring the lizards to an indoor facility for 32 lizards (16 males and 16 females), while the remaining 32 lizards (16 males, 16 females) were temporarily held in small holding bins until we finished the first batch of lizards; owing to logistical and space constraints we were only able to process 32 lizards at a time. All lizards were provided with a middle refuge and had constant access to water and UV lighting during the experiments. Heat cord (30°C) was used to create a thermal gradient in each enclosure so that lizards had ample opportunity to thermoregulate. Crickets were fed ad libitum every second day except during the first 6 days of experiments because we did not want lizards to be satiated during feeding trials with novel prey (details in §2(d)i below).

(b) Behavioural assays

All behavioural trials described below were conducted in the lizards' own holding enclosures, recorded using a mounted security camera (Swann security system), and later scored in JWatcher (http://www.jwatcher.ucla.edu/). Video footage was scored 'blind' to whether individuals had learnt the spatial test, and subsets of data (e.g. trials within each experiment) were scored by the same individual to avoid inter-observer bias [43]. Full details of all behavioural assays can be found in the electronic supplementary material.

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(c) Exploratory behaviour (day 1)

Lizards were introduced into a novel enclosure (683 (L) × 483 (W) × 385 (H) mm) with two weighted black refuge-boxes (170 × 65 × 120 mm). Each refuge had three entrances (front and sides) and was placed at each end of the enclosure (electronic supplementary material, figure S1). After 3 min acclimation in the central refuge-box, the refuge was lifted and trials were run for 30 min. We scored the following behaviours: (i) time in locomotion (TL) and (ii) time taken to enter the two refuges in the enclosure (T2ER). Lizards that did not visit both refuges within 30 min were assigned a latency of 1800 s. We included the latency to visit both refuges as a measure of quickness to explore in a novel environment, and locomotion as a measure of the amount of exploration.

(d) Measures of boldness

Boldness is most often interpreted as the tendency to take risks, especially in novel situations, and is usually measured experimentally in relation to anti-predatory behaviours or individual responses to novel cues [44]. In this study, we used two experiments to assay boldness separately in an anti-predatory and in a neophobia context.

(i) Assay I: neophilia (days 2–7)

We quantified variation in neophobia/neophilia by examining individual lizard responses when presented with a novel food item (i.e. a dead silkworm pupa). We presented the pupa by gently dangling it in front of the lizard (or at the entrance of the refuge it was in) for 3 min (once per day for five consecutive days). On day 6, we presented each lizard with a pupa left hanging *ca* 1–2 cm from the centre of the tub for 1 h, and in the absence of observers. Lizards were divided in two categories (NEO): neophilic (lizards that ate the novel prey at some point during trials) and neophobic (lizards that did not eat the novel prey at all).

(ii) Assay II: anti-predatory trial (day 8)

Assays began by gently chasing each lizard into the middle refuge and randomly designating one of the two lateral refuges as the 'hot' refuge (by suspending a 60/100 W incandescent bulb *ca* 25 cm above this refuge) and the other as the 'cold' refuge (by packing a box with ice and placing it beneath the refuge, under the tub). We then removed the central refuge, left lizards to acclimate for 5 min, switched the basking light over the 'hot' refuge on and allowed lizards 15 min to reach the basking platform and initiate basking. We finally simulated a predatory attack by chasing the lizard off the basking refuge until it entered into the 'cold' refuge at the opposite end of the tub. After the simulated attack, we measured the time it took lizards to return to their basking sites (LATB; 45 min maximum).

(e) Spatial learning trials (days 9–28)

To measure spatial learning, we set up a simple spatial learning assay using an anti-predatory paradigm that has been used successfully in previous studies [32,34]. We initiated trials by re-introducing the two side refuges in the same positions as in the anti-predatory trial (the 'cold' refuge in the anti-predatory trial was selected to act as the 'safe' refuge across spatial learning trials) and removing the middle refuge, after which lizards were given a variable amount of exploration time (30–45 min). After this time, we scared lizards around the enclosure until they entered the 'safe' refuge. If the lizard entered the 'unsafe' refuge, then we lifted the refuge and resumed chasing the lizard until it entered the 'safe' refuge. Each lizard was tested once a day for an overall period of 20 days (i.e. 20 trials). We measured whether it chose the correct 'safe' refuge or not ('choice'), the number of extra incorrect choices (i.e. number of choices after the first incorrect choice) needed to choose the correct refuge ('incorrect choices'), and the overall latency to enter the 'safe' refuge ('latency'). The rationale for choosing these variables was: (i) to keep the variable 'incorrect choices' independent of the variable 'choice' and (ii) because 'rule-of-thumb' learning of the type 'run into a refuge and if chased again run straight into the other one' could explain a significant drop in both 'incorrect choices' and 'latency' across trials without the need to invoke spatial learning. Hence, we analysed learning curves for these three variables. Owing to variation in the time of the day in which trials were conducted, which was balanced within individuals every four days (see the electronic supplementary material for details), results were blocked every four days (i.e. four trials). Lizards were categorized as learners or non-learners according to their cumulative tally of correct/incorrect choices across the 20 trials. A lizard was considered to have chosen correctly when it was already found inside the 'safe' refuge at scaring time or when the first refuge it ran into in response to the simulated predatory attack was the 'safe' refuge. We considered a lizard to have learnt when: (i) following at least four consecutive correct choices it accumulated a significant correct/incorrect tally according to a binomial distribution (e.g. 5/5, 7/8, etc.) and (ii) when, from this point on, its overall correct/incorrect tally until the end of trials remained significant (see the electronic supplementary material for full details about behavioural trials and data on correct/incorrect tallies for all lizards across the 20 trials).

(f) Statistical analyses

We were able to obtain learning data from 64 lizards (but conservatively excluded two ambiguous non-learners from formal analyses because they exhibited a higher proportion of wrong choices than expected by chance), and full behavioural, learning and morphological data from 56 lizards (owing to missing data; see the electronic supplementary material for details). We used generalized estimating equations (GEEs) to analyse learning curves for our 'choice', 'incorrect choices' and 'latency' variables ($n = 31$ per sex). We rounded latency to the nearest whole number and modelled both latency and number of incorrect choices using a Poisson error distribution, and lizard choices using a binomial error distribution. We included lizard ID as the grouping variable and used an autoregressive 1 correlation structure (AR1). The GEEs estimate a scale parameter and account for over-dispersion in the models. We compared models using Wald tests to test for significant block, sex and sex × block effects as well as significant effects of learning, block and learning × block effects.

To analyse the relationship between behavioural traits and learning, we used generalized linear models (GLMs) with a binomial error distribution and 'logit' link function, in R v. 2.14.0 [45]. Learning (binary: 'learn' or 'no learn') was modelled as a function of sex and body condition, exploratory behaviour (i.e. time to explore two refuges (T2ER) and time moving in novel environment (TL)) and boldness (i.e. latency to bask after simulated predatory attack (LATB), and whether an individual ate a novel prey item (1 = ate; 0 = did not eat) (NEO)). Body condition was calculated as the scaled mass index [46], but note that using the residuals from a linear regression between log mass and log body size (SVL) [47] yielded equivalent results. Using the latter independent variables and sex and condition as covariates, we generated a series of candidate models based on *a priori* hypotheses about the role that behavioural types play in affecting the probability of learning based on previous results from birds and fish [12–14,48]. Graphical inspection of variables suggested that LATB was not necessarily linearly related to the probability of learning so we included a quadratic parameter in models with LATB. Given our sample size, we limited the number of

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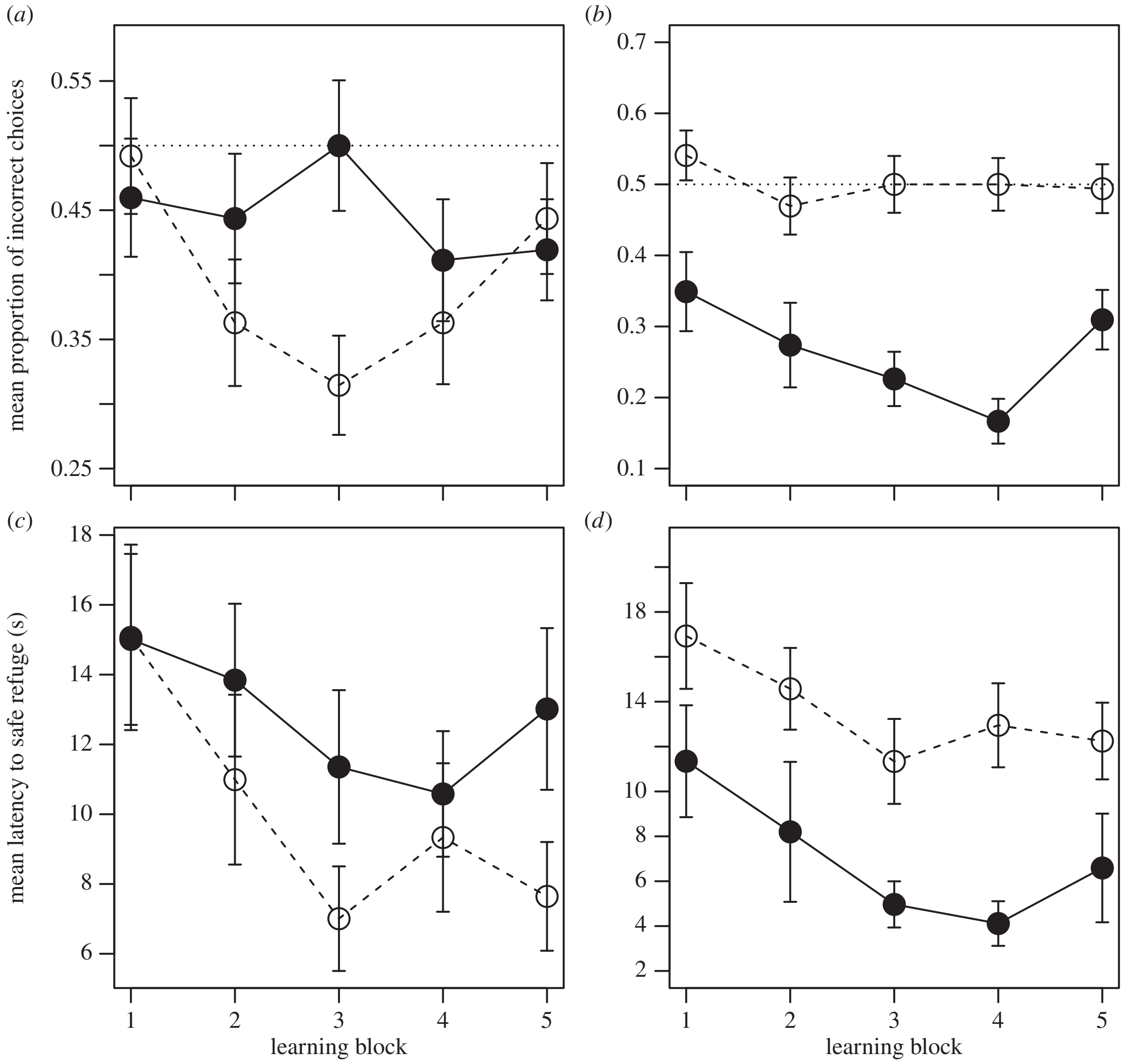

Figure 1. Learning curves for males (white circles in *a,c*; $n = 31$) versus females (black circles in *a,c*; $n = 31$); and learner (black circles in *b,d*; $n = 21$) versus non-learner lizards (white circles in *b,d*; $n = 41$). The dashed line in (*b,c*) marks the probability of entering into the correct refuge by chance given the experimental set-up. Note that instances in which lizards were already encountered in the 'safe' refuge at the time of the simulated attack (see Material and methods) were not considered for the analysis of latencies.

parameters to a maximum of five and did not include both measures of exploration in the same model to avoid possible autocorrelation. Prior to analysis, we standardized our independent (input) variables (mean = 0, s.d. = 2) to allow interpretation of main effects in the presence of higher order parameters and to ease comparisons among model estimates.

Alternative models were evaluated using the second-order information criterion, AIC_C, owing to a parameters-to-sample-size ratio of less than 40 [49]. As there was no clear 'best' model in our candidate set (i.e. Akaike model weight greater than 90%), we adopted a model averaging approach in addition to presenting our top-ranked model [50]. We chose to present model-averaged coefficients because we made predictions about the direction of individual parameter estimates and because the hypothesized role of behavioural types on learning is still in its infancy, and hence we feel that it is important to present effect sizes for all hypothesized parameters to guide future research [50]. We also used our top model and the estimated parameters to make predictions about the probability of learning given that the coefficients in this top model explain the greatest amount of variation in our data. We did not exclude models in the candidate set that were more complex versions of reduced models during model averaging [49] to avoid exclusion of biologically relevant effects. Owing to our limited set of models and hypothesized relationships between learning and our behavioural traits, we chose natural model averaging using the models that had a cumulative model weight of 95% [49]. We inspected the fit of all top models (2 ΔIC) by looking for influential points (Cook's distance and hat values) and testing for colinearity using variance inflation factors. We found weak evidence for over-dispersion (residual deviance/residual degrees of freedom less than or equal to 1.3). The model with the highest dispersion was the top model; however, re-fitting this model with a quasi-binomial error distribution, where a dispersion parameter is estimated, did not affect the results.

3. Results

(a) Sex differences in spatial learning

Twice as many males learnt the spatial task within 20 trials compared to females (14 of 31 (45.1%) males and 7 of 31 (22.6%) females). The analysis of learning curves revealed a significant sex × block interaction in the probability of choosing the correct refuge ($\chi^2 = 11.1$, $p = 0.026$). Males where significantly more likely to choose the correct refuge by Learning Block 3 (estimate $= -0.90 \pm 0.33$, $Z = 7.32$, $p = 0.007$). This difference disappeared in blocks 4 and 5, by which time females were already performing below chance (i.e. had learnt the task; figure 1), so our results strongly suggest that males were quicker

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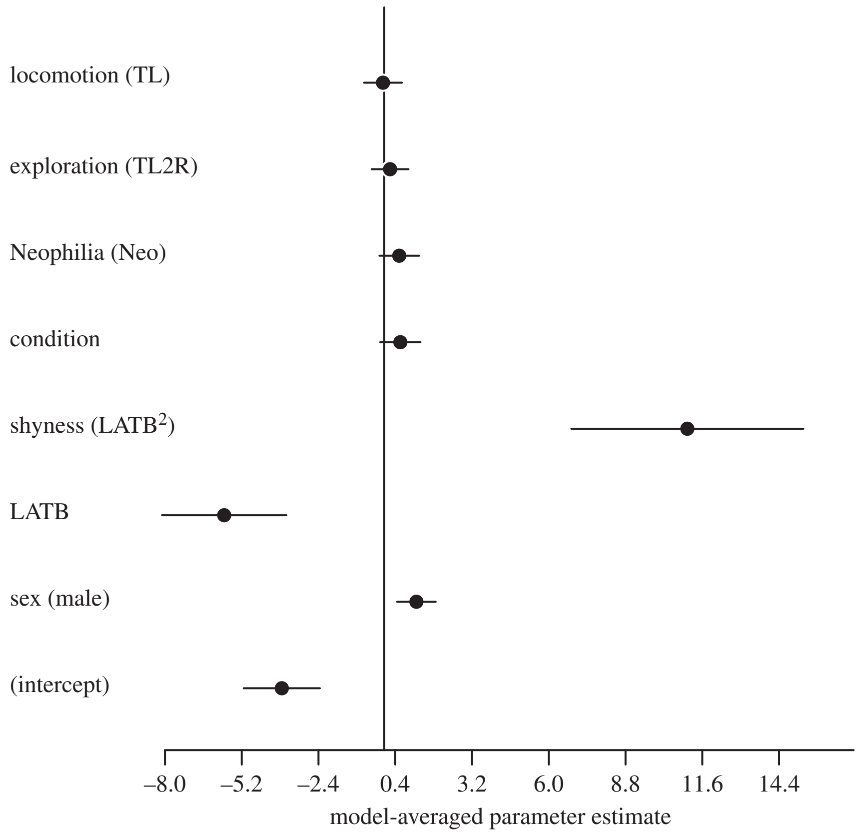

Figure 2. Parameter estimates after natural model averaging of models with a cumulative model weight of 95% (see Material and methods).

than females in learning the spatial task. The number of extra incorrect choices needed to find the safe refuge decreased significantly across blocks (Wald $\chi^2 = 17.3$, $p = 0.002$; figure 1*a*), but females did not show a clear tendency to make a higher number of incorrect choices than males (Wald $\chi^2 = 2.24$, $p = 0.13$; figure 1*a*), with no evidence for a significant sex × block interaction (Wald $\chi^2 = 2.63$, $p = 0.62$). Similarly, males and females did not differ significantly in their latency to enter the safe refuge (LAT) across blocks (sex × block: Wald $\chi^2 = 3.4$, $p = 0.49$; sex: Wald $\chi^2 = 0.312$, $p = 0.58$; block: Wald $\chi^2 = 4.54$, $p = 0.34$; figure 1*c*), which shows that sex-differences in learning cannot be attributed to differences in rule-of-thumb learning. Males exhibited a notable upward swing in the probability of choosing the incorrect refuge in the last learning block. This could be due to overtraining effects on attention [51] or, perhaps more likely, to gradual extinction of the efficacy of the negative reinforcement across trials due to lizards habituating to human presence/scaring.

(b) Individual behavioural type and the probability of spatial learning

Our two measures of exploratory behaviour, TL and T2ER, explained little variation in the probability of learning, with all models containing these two variables being more than 2 ΔAIC_C units from the top model (electronic supplementary material, table S1; Model 11 and Model 12). This was also evident in the model-averaged estimates with these two variables having small effect sizes; TL2R had a positive estimate while TL had a slightly negative estimate (figure 2).

Models that included whether a lizard ate a novel food item or not (NEO) approached 2 ΔAIC_C units from the top model (Model 10; electronic supplementary material, table S1) and the model-averaged estimate showed a small positive effect

Table 1. Parameter estimates and 95% CI around estimates for the top-supported model (Model 8; electronic supplementary material, table S1). Note that coefficients are standardized $[(x_i - \mu_i)/(2 \times \sigma_i)]$.

coefficient	estimate	lower 95% CI	upper 95% CI
intercept	−3.79	−6.70	−1.66
sex (male)	1.20	−0.09	2.63
LATB	−5.82	−10.71	−2.07
LATB^2	10.98	4.02	20.18

on the probability of learning (figure 2). Models containing only the main effect of the latency to return to the basking refuge after a predatory attack (LATB) poorly explained variation in the probability of learning (electronic supplementary material, table S1; $\Delta AIC_C > 10$); however, there was evidence that the relationship between the probability of learning and LATB was nonlinear (table 1 and electronic supplementary material, table S1). The best-supported model contained sex, LATB and LATB^2 (electronic supplementary material, table S1) and confirmed that LATB and LATB^2 had significant effects on the probability of learning (table 1 and figure 2). The predicted probabilities of learning showed that there were two groups of individuals with a high probability of learning located at the extremes of this distribution (figure 3). Individuals with short LATB ('bold') had a high probability of learning the task and there was a sharp decline in this probability of learning to individuals with intermediate latencies (figure 3). The probability of learning the spatial task increased again for individuals taking a long time to return to the basking

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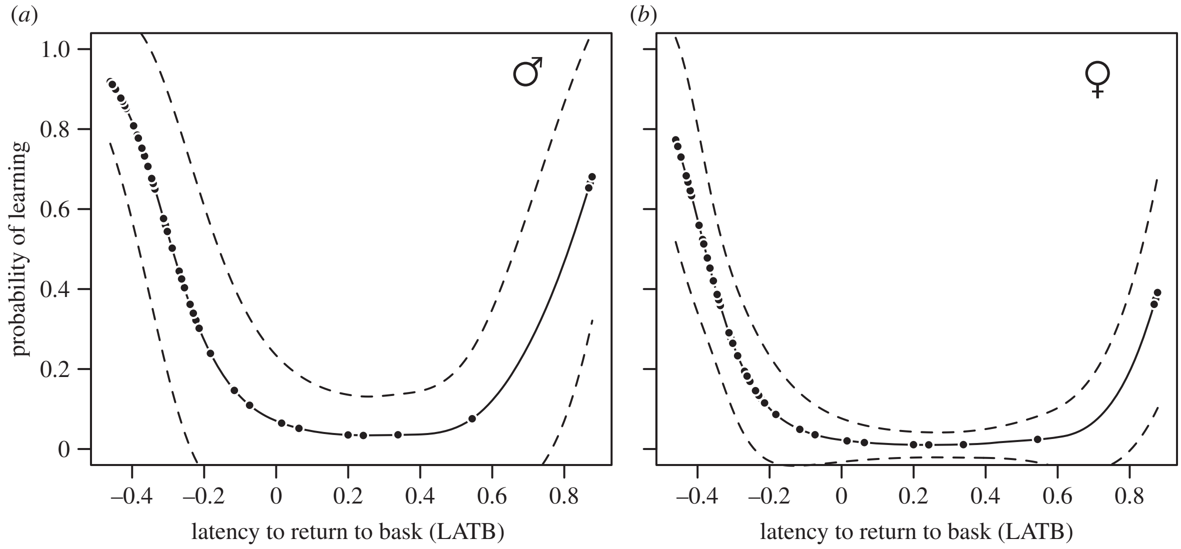

Figure 3. The predicted probability of learning a spatial task for (*a*) males and (*b*) females as a function of the latency to return to a basking refuge after a simulated predatory attack (LATB). Predicted probabilities are based on our top-supported model (Model 8; electronic supplementary material, table S1) using standardized input variables $[(x_{\mathrm{LATB}_i} - \mu_{\mathrm{LATB}})/(2 \times \sigma_{\mathrm{LATB}})]$. We used our standardized input LATB variable to predict probabilities and fit a smoothed cubic spline function to the data. Dashed lines above and below fitted lines are the 95% CIs (predicted (p) $\pm$ 1.96 $\times$ s.e. of fit and smoothed with a cubic spline).

refuge ('shy'). 'Bold' males (i.e. individuals at −0.4 units from the mean) were predicted to have an 82% probability of learning, while 'bold' females had a 57% probability of learning the spatial task (figure 3*a*,*b*). By contrast, 'shy' males (individuals at 0.8 units from the mean) were predicted to have a 45% probability of learning the spatial task while 'shy' females were predicted to have a 20% probability of learning (figure 3*a*,*b*).

4. Discussion

Our results strongly suggest that male *E. quoyii* performed significantly better and faster at our spatial learning task than females. We also found significant variation among individuals in boldness (latency to exit a refuge and return to a basking platform after a simulated attack), and a significant association between variation in boldness and the probability of learning a spatial task: both 'bold' and 'shy' behavioural types were more likely to learn the spatial task than individuals with intermediate behaviour.

(a) Sex differences in spatial cognition

Sex differences in cognitive ability in mammals and other taxa are most commonly documented for spatial cognition, where males typically perform better than females [1,52–54]. Sex differences in spatial cognition have been hypothesized to arise from differential selective pressures in relation to sex-specific dispersal, mobility during reproduction, intrasexual competition, female choice and differences in home range size (i.e. the range-size hypothesis [31]). Although the latter hypothesis seems to have the most support [25,52], most hypotheses actually link spatial ability to space use and differ only in their explanations as to why the sexes differ in their use of space.

In many lizard species, reproduction seems to pose higher spatial challenges to males compared with females. In particular, males generally possess larger home ranges and/or need to process more complex spatial information than females in order to achieve copulations (e.g. home-range boundaries, location of rivals, location of females within their home range [25,35,36]). As predicted by such differences between male and female social roles [55], we found that male *E. quoyii* were better at our spatial learning task than females. More males successfully learnt the spatial learning task than females (45.1% male learners versus only 22.6% female learners), and our results also suggest that males were quicker at learning the spatial task than females. To the best of our knowledge, sex differences in learning have never been reported in a reptile, which is particularly striking in the context of lizard spatial cognition because sexual differences in spatial demands seem widespread in lizards [55]. Our study is therefore the first evidence to date for sexually dimorphic cognitive performance in a reptile. Although we used only one learning task to gauge spatial learning, and hence our results cannot be extrapolated to spatial learning in other contexts, we suggest that the home range size hypothesis put forward to explain sexual differences in spatial cognition in mammals may similarly apply to lizard species in other spatial learning contexts [25]. Future studies should address this hypothesis, and the generality of the findings reported here.

(b) Boldness, spatial learning and alternative reproductive tactics in *Eulamprus quoyii*

We found considerable variation in individual latency to return to basking after a simulated predatory attack, and a clear association between variation in this boldness measure and variation in individual spatial learning performance. 'Bold' individuals (quick to return to bask after attack) had the highest probability of learning the spatial task. However, extremely 'shy' behavioural types also learnt the spatial task with a higher probability than individuals with intermediate behavioural types. This is the first evidence linking cognition and behavioural types in a reptile, and it is inconsistent with

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the risk–reward hypothesis and with previous studies [4,21,56,57], where the relationship between boldness and learning was always reported to be linear.

We suggest a tentative but interesting possibility that both extremely 'bold' and 'shy' behavioural types may have enhanced spatial learning because of the ARTs they adopt. ARTs in *Eulamprus* are associated with divergent selection for different behavioural types that are likely part of a territorial-floater behavioural syndrome relating to boldness, activity and exploratory behaviour, as has been shown in this genus and in other lizards [35,40,41]. Typically, territorial lizards actively defend against other males core areas that tend to overlap the home range of several resident females, while 'floater' males instead navigate their way over larger areas, traversing several different territories in their search for copulations with females. In *Eulamprus*, territorial males are bolder, more active and explore less than floater males, which matches the behaviour predicted by their ART but does not conform to the typical fast–slow, high–low, risk–reward trade-off behavioural axis [39–41]. Interestingly, the fitness of each of these ARTs probably depends greatly on spatial cognition. Territorial males need to process and memorize detailed spatial information (e.g. the position of females within a territory, territory boundaries with neighbouring males [36,58]). Similarly, floater males need to navigate their way over large home ranges consisting of varied habitat, and where the locations of refuge sites, rival male territories and potential mates are important for both reduced conflict and reproduction, while also aiding survival [35,39]. Under this scenario, divergent behavioural types may have given rise (via divergent correlational selection or developmental pathways) to enhanced spatial learning for both of these male reproductive tactics. ARTs have also been suggested in female *Eulamprus* in relation to territory residency, anti-predatory behaviour and exploration [41], although to date there is no direct evidence of a clear link between these behavioural traits in *E. quoyii* females. Alternatively, strong selection for divergent cognitive–behavioural types in males may also have led to correlated selection in females [59]. Indeed, there does appear to be strong sexual selection on the behavioural traits associated with each of the ARTs [39].

As a word of caution, and while the above is certainly compelling, our results cannot be taken as direct support for this hypothesis. First, while ARTs have been documented in *Eulamprus*, we did not directly assess the ARTs of the individuals used in our study. Second, we did not find any evidence that exploration and/or boldness in the neophilia experiment were significantly associated with spatial learning, which is something that we would have expected if ARTs were shaping cognitive–behavioural syndromes. In particular, we would have expected neophilia/boldness and slow exploration (i.e. characteristic of territorial males) and neophobia/shyness and fast exploration (i.e. characteristic of floater males) to be positively associated with spatial learning in male lizards. Plotting these two variables separately for male learners versus non-learners (*post hoc*) does hint at such a relationship in males (electronic supplementary material, figure S2), but this was not picked up in our analysis (perhaps owing to the small sample sizes). We suggest that future studies should measure ARTs (i.e. territorial behaviour) directly in the wild, and then relate this to spatial learning as measured in the laboratory or, ideally, in the field.

5. Conclusion

To conclude, we provide the first evidence of sex-dependent spatial learning in a reptile. We suggest that sexual dimorphism in spatial learning may be linked to the different social roles (and ensuing spatial demands) experienced by males and females in territorial lizards and that consequently this phenomenon may be more widespread than previously suspected. We also show that both 'bold' and 'shy' behavioural types have enhanced spatial learning and propose that this may be because of their association with ARTs in *E. quoyii*. This is also the first evidence that behavioural traits such as boldness are associated with learning in a reptile and, along with recent studies in birds, highlights the importance of considering cognitive traits in the study of behavioural syndromes, and *vice versa*. We suggest that future studies consider different social roles and tactics as an important factor in the evolution and/or development of specific behavioural–cognitive syndromes. In lizards, a first step to test this hypothesis would be to examine the link between territorial behaviour (i.e. ARTs), spatial learning and fitness. Characterizing whether spatial learning is under strong selection and, if so, examining its relative strength across different social roles and tactics are bound to provide crucial insight into our understanding of how intraspecific variability in spatial cognition arises.

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Data accessibility. Behavioural data deposited in dryad: doi:10.5061/dryad.j83c0.

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Online Supplementary Material

Materials and Methods*

**Please note that in order to make the methods as clear as possible we have repeated some information from the manuscript.*

Exploratory Behaviour (Day 1)

We removed lizards from their holding tubs and transferred them to the lab, where they were released into a novel enclosure measuring 683 (L) x 483 (W) x 385 (H) mm. Each enclosure had paper substrate that was gridded with 5 x 5 cm squares. Two weighted black refuge boxes (170 x 120 x 50 mm) with three entrances each were situated at each end of the enclosure (Fig. S1). Immediately before trials, we placed lizards under a central acclimation refuge (identical to enclosure refuges) that we lifted from behind a blind after 3 minutes. Trials were initiated at this moment and lasted for 1800 s.

Assay I – Neophilia (Days 2-7)

We inserted a third refuge against the side of one of the long-sided walls of the tub, in a central position, with the entrances facing inwards (Fig. S1). To obtain a neophobia/neophilia estimate, we examined individual lizard responses when presented with a completely novel food item. We used commercially available dead silkworm pupae, which are often used as a supplement food for reptiles but are unavailable in their natural habitat. We selected silkworm pupae not only because they constitute a novel visual and chemical stimulus, but also because we could standardize their movement during presentation. We suspended a single pupa from the end of a transparent fishing line threaded through the body cavity. The fishing line was tied to a wooden dowel, which the researcher controlled from behind a blind. Immediately preceding trials, we moved the focal lizard enclosure to the recording area and made note of the location of each lizard within its enclosure. Lizards were then given between 2 - 5 minutes to acclimatise before starting trails. During trials, we presented the pupa by gently dangling it in front of the lizard for 3 minutes. Pupae were presented at ca. 15 cm (i.e. three 5 cm squares) from the lizard's snout for the first minute and this distance was decreased by 5 cm each minute thereafter. All trials were recorded. After the first neophobia trial, we repeated this procedure once a day for four more consecutive days (five days total). Finally, on day 6 we presented each lizard with a pupa that was suspended in the same manner as in prior feeding trials, but which we left hanging ca. 1 - 2 cm from the centre of the tub for one hour, and in the absence of observers. We recorded at which stage during these trials each lizard first attacked or ate the novel prey item. Only 25 of 64 (39%) lizards actually consumed the larva, although lizards that ate did so repeatedly throughout the feeding trials. We therefore divided lizards in two categories (NEO): neophilic (lizards that ate the novel prey at some point during trials) and neophobic (lizards that did not eat the novel prey at all).

Assay II – Anti-predatory trial (Day 8)

Lizards were tested in their own holding tub, which was no longer a 'novel' environment. The assay began by gently chasing each lizard into the central refuge (Fig. S1). We then randomly selected one of the two remaining refuges and designated one as the 'hot' refuge by suspending a 75 W incandescent bulb ca. 25

cm above this refuge, effectively transforming the top surface into a basking platform. Finally, we packed a box with ice and placed it beneath the second refuge, under the tub (i.e. the 'cold' refuge; Fig. S1). During anti-predatory trials, ambient room temperatures were cooled down (22 – 24°C) to promote basking by lizards. Once the assay had started, we removed the central refuge and left lizards for five minutes to acclimate, after which time we switched the basking light on and allowed lizards 15 minutes to reach the basking platform and initiate basking. Once the 15-minute period had elapsed, we simulated a predatory attack by chasing each lizard off the basking 'hot' refuge until it entered into the 'cold' refuge situated at the opposite end of the tub. Researchers simulating a predatory attack wore blue nitrile gloves and used their fingers to chase lizards around in a standardized way (i.e. always from behind by gently tapping lizards on the base of their tail). Simulated predation by an approaching human has been used extensively in the past for lizards because they are known to respond to an approaching human as a predator (Font et al. 2012. Journal of Comparative Psychology, 126: 87-96). After simulated attacks, lizards were allowed 45 minutes to emerge from the 'cold' refuge and return to the basking platform, during which time we measured the time to return to their basking sites (LATB).

Spatial learning trials (days 9-28)

Following anti-predatory trials on day 8 we: i) reintroduced the middle refuge (Fig. S1); ii) removed the 'hot' and 'cold' refuges along with the basking bulb and ice-box; and iii) introduced a distinct 'intra-maze' cue consisting of a piece of green rectangular cardboard pasted to the wall opposite to the middle refuge; this served to disambiguate spatial orientation during trials. Lizards were tested in their own holding tubs, which were transferred to the filming area just before the beginning of trials. Each lizard tub was positioned in the same position and orientation within the room to keep extra-maze cues constant across trials. When moved to the filming area, lizards invariably entered the middle refuge (the only one available at this time). We took advantage of this circumstance to suspend an incandescent bulb from the middle of the tub and introduce two refuges in the same positions as the 'hot' and 'cold' refuges used in the anti-predatory trial (Fig. S1). For each lizard, the refuge positioned where the 'cold' refuge had been in the anti-predatory trial was selected to act as the 'safe' refuge in the spatial cognition trials. Hence, the position of the 'safe' refuge was randomly determined for each individual, but was kept constant across trials. We initiated trials by removing the middle refuge, after which lizards were given a variable amount of time to explore the tub with the 'safe' and 'unsafe' refuges in place. We randomly varied the time (30 – 45 minutes) lizards were given to explore before simulated predatory attacks to prevent them from learning to anticipate attacks. After this time, we followed the scaring protocol used in the anti-predatory trial to simulate a predatory attack. We scared lizards around the tub and, when necessary, out of the 'unsafe' refuge until they entered into the 'safe' refuge. Once a lizard entered the 'safe' refuge, we removed the 'unsafe' refuge and we put the central refuge back into the enclosure. 'Safe' refuges were left inside the enclosure until dusk, at which time they were gently removed from the tub taking care not to startle lizards that might be inside them. Note that we always used the same two lateral refuges for each lizard, but randomized its use as safe and unsafe refuges across trials to remove the possibility that lizards were using chemical cues to locate the safe refuge. Each lizard was tested once a day for an overall period of 20 days (i.e. 20 trials). For each lizard, we measured whether it chose the correct 'safe' refuge or not, the number of extra incorrect choices (i.e. number of times it ran into the 'unsafe' refuge), and the overall latency to enter the 'safe' refuge. A lizard was considered to have chosen correctly when it was already found inside the 'safe' refuge at scaring time or when the first refuge it ran into in response to the simulated predatory attack was the 'safe' refuge. Cognition trials were conducted at four different time slots each day: morning (10:00 – 11:30), midday (11:30 – 13:00), afternoon (13:00 – 14:30) and early evening (14:30 – 16:00). We divided the 32 lizards in each batch into four groups of eight lizards and scheduled trials so that we counter-balanced the time of the day in which lizards were tested. Hence, each lizard was tested at a different time each day and went through all the time slots every four days; consequently, when plotting learning curves, results were blocked every 4 days (i.e. 4 trials) into a single learning block so that each lizard was subject to 5 learning blocks overall. We assigned lizards as learners using a two-step procedure based on the tally of correct versus incorrect choices over the 20 consecutive trials. A lizard was considered to have chosen correctly when it was already found inside the 'safe' refuge at scaring time or when the first refuge it ran into in response to the simulated predatory attack was the 'safe' refuge. First, we examined the tally of correct/incorrect choices (i.e. 20 per lizard) chronologically and marked those lizards that exhibited a tally of consecutive correct/incorrect choices that was significant according to a one-way binomial test (e.g. 5 over 5, 7 over 8 or 9 over 11 consecutive correct choices), and that begun with four consecutive correct choices (i.e. a 7 over 8 tally that did not start with at least 4 consecutive correct choices was not considered), as potential learners. Second, we also examined the overall tally of correct/incorrect choices from the first trial in the significant tally to the end of testing and only considered lizards as learners if this tally was also significant. For example, a lizard that made 5 consecutive correct choices in trials 3 to 7 was not considered as a learner if it made more than 4 incorrect choices from trial 7 to the end of testing in trial 20 (i.e. if its overall correct/incorrect tally from trial 3 to trial 20 was lower than 13 correct choices over the 17 trials). We implemented this second criterion to avoid categorising lizards as learners by chance (see Table S2 for raw data). According to our criterion, 7 lizards that were categorized as potential learners (i.e. step one in our criterion) were finally categorized as non-learners based on the requirement that the binomial tally remain significant from the first trial in the first significant tally until the last trial (see Table S2). Five of these lizards ended up with correct/incorrect tallies that were clearly ($p > 0.1$) non-significant (female 6: 12/19, female 219: 10/16, male 55: 6/8, male 155: 14/20, male 36: 9/13), strongly suggesting that they were correctly categorized as non-learners. However, male 158 (9/12; $p = 0.073$) and female 3 (11/15; $p = 0.059$) did exhibit an overall tally that was closer to being significant. To ensure that our results were not affected by imposing a learning criterion that was not too strict, we re-ran our analysis including these two lizards as learners. The results were equivalent to those reported using our original categorization of learners (Table S3;

Estimates, mean ± se, for the top supported model: Sex males = 1.07 ± 0.65, LATB = -5.05 ± 1.99, LATB 2 = 9.88 ± 3.70; Model-averaged estimates, mean ± adjusted se: Sex males = 1.05 ± 0.65, LATB = -5.09 ± 2.03, LATB2 = 10.03 ± 3.75, Neo = 0.63 ± 0.71, body condition = 0.57 ± 0.69, L2R = -0.31 ± 0.66, L = 0.24 ± 0.67)..

Finally, and as noted in the manuscript, due to an unexpected contingency (camera malfunction) we missed considerable behavioural data from 3 males and 2 females, and morphological data (due to a Pit-tag loss) from one female, so we excluded these lizards from our analyses of the relationship between behavioural traits and learning (n=56). Behavioural data could not be recovered, but we did re-run these models including the one female for which we had lost the morphological measurements by inputting female scaled body mass through mean imputation (i.e. replacing her missing condition index with the mean condition index). This analysis was qualitatively identical to the analyses reported in the main manuscript (Estimates, mean ± se, for the top supported model n = 57: Sex males = 1.01 ± 0.65; LATB = -5.36 ± 1.88; LATB^2 = 11.36 ± 4.14)

Fig. S1 - Layout of lizards' experimental/holding tubs during each of the four experiments. *a)* Experiment 1 consisted of a single 30 min trial (day 1). Immediately before this trial, two refuges (170 x 65 x 120 mm; R1 & R2) were introduced in the tub. At the end of this experiment, we added a third refuge (Fig.1*b*; MR), which was to remain as a permanent 'home refuge' throughout the rest of the experiments (except during some trials, see below). *b)* Experiment 2 consisted of 5 trials (days 2-6). Lizards had constant access to the three refuges (R1, R2, MR) throughout this experiment, but R1 and R2 were removed when it ended. *c)* Experiment 3 consisted of a single 45 min trial (day 8). Immediately before starting, we introduced a 'hot refuge' (HR) under a basking light and a 'cold refuge' (CR). 'Cold refuges' were cooled down by sliding a packet of ice underneath the tub (see methods). Note that at this time all lizards were inside their home refuge (MR). Trials began by removing MR, after which lizards were left alone for 15 min. After the 15 min had elapsed, all lizards were basking on top of the HR and we immediately chased them into CR. We terminated trials once a lizard resumed basking, or after 30 min. At the end of experiment 3, we removed the HR and CR, reintroduced the MR, and pasted a green rectangular piece of cardboard (IMC; Fig. 1d) on the wall opposite to MR to disambiguate the spatial arrangement of tubs during spatial learning trials. *d)* Experiment 4 consisted of 20 trials (see methods). Immediately before each trial, we shuffled the mulch around to spread chemical cues (except the one underneath MR) and introduced the two new refuges (i.e. 'safe' (SR) and 'unsafe' (UR) refuges). We started trials by removing the MR. At the end of each trial, we reintroduced the

MR and removed the UR. The SR was left inside the tub until lizards switched back to the MR or until the lights were off at the end of the day (at which time it was gently removed).

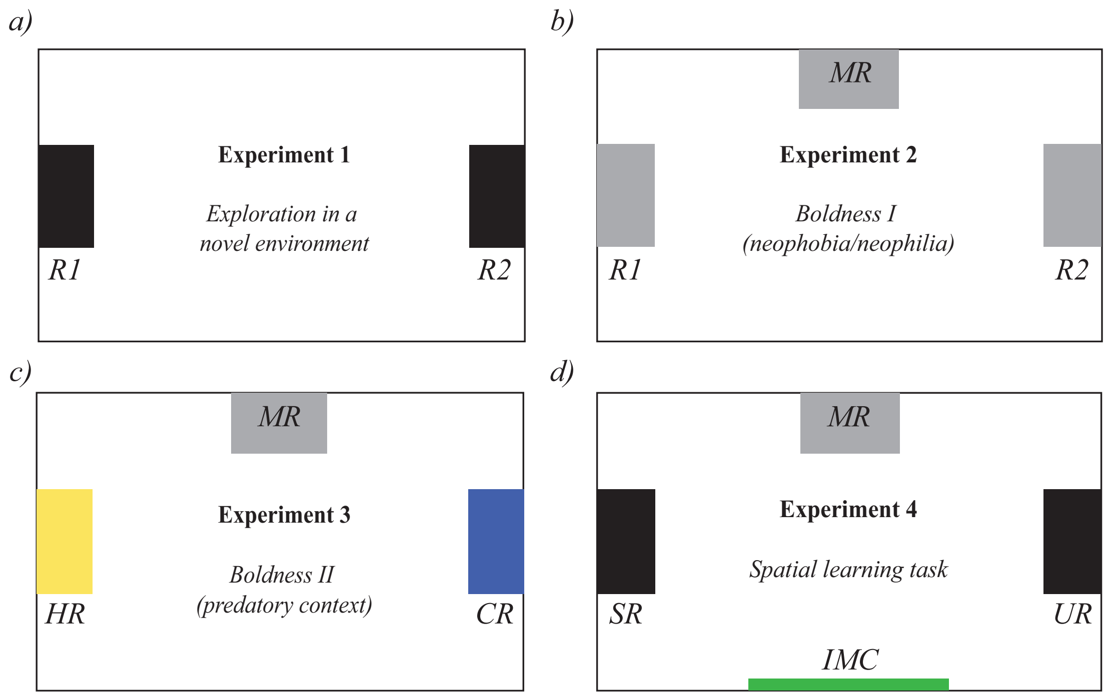
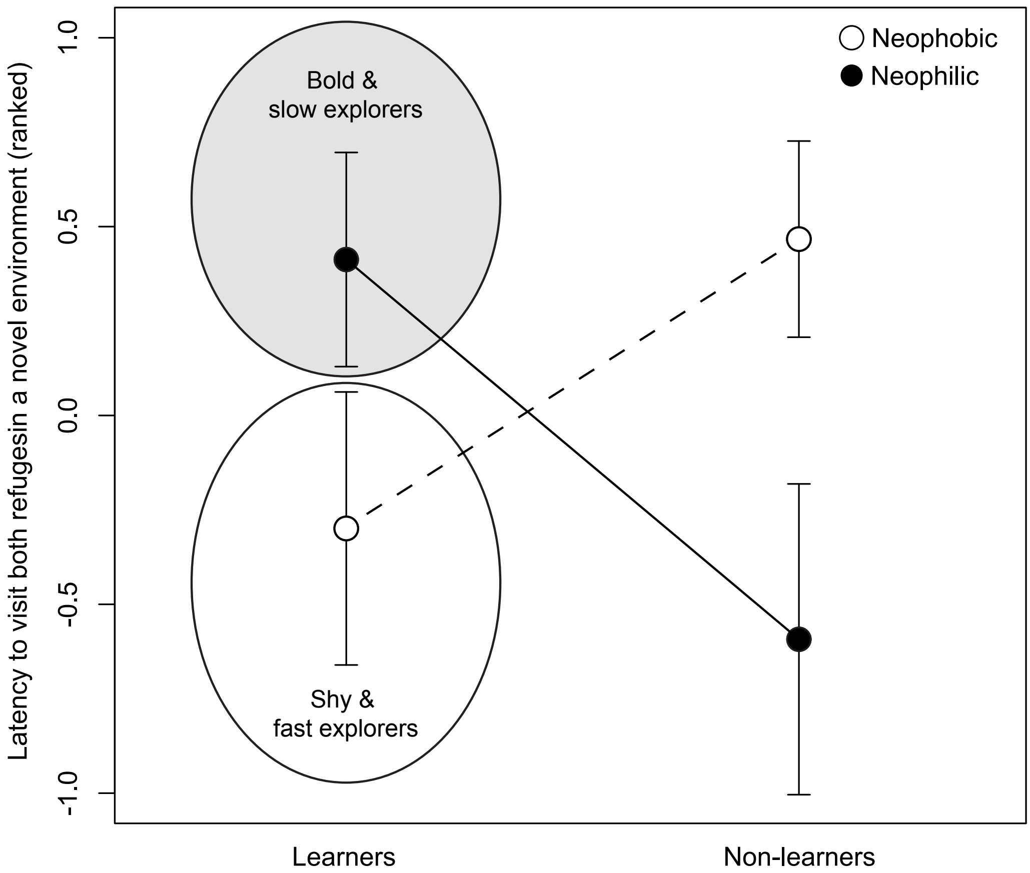


Fig. S2- Interaction plot looking at male differences in exploration (T2ER) depending on whether male lizards learnt or not (learners vs. non-learners), and ate or not in the food neophobia experiment (NEO). T2ER shows the ranks converted to normal quantiles to normalize the rank data. Existing evidence suggests that territorial males are bold and slow explorers in a novel environment while floater males are shy and fast explorers in a novel environment (Stapley & Keough 2004, 2005; Noble et al. 2013).



Electronic supplementary material for: Carazo, P., Noble, D. W. A., Chandrasoma, D. & Whiting, M. J. 2014. Sex and boldness explain individual differences in spatial learning in a lizard. *Proc R Soc B*

Figure S3. Individual learning curves looking at the proportion of trials in which lizards chose the incorrect refuge across the 5 learning blocks in (a) non-learners, and (b) learners.

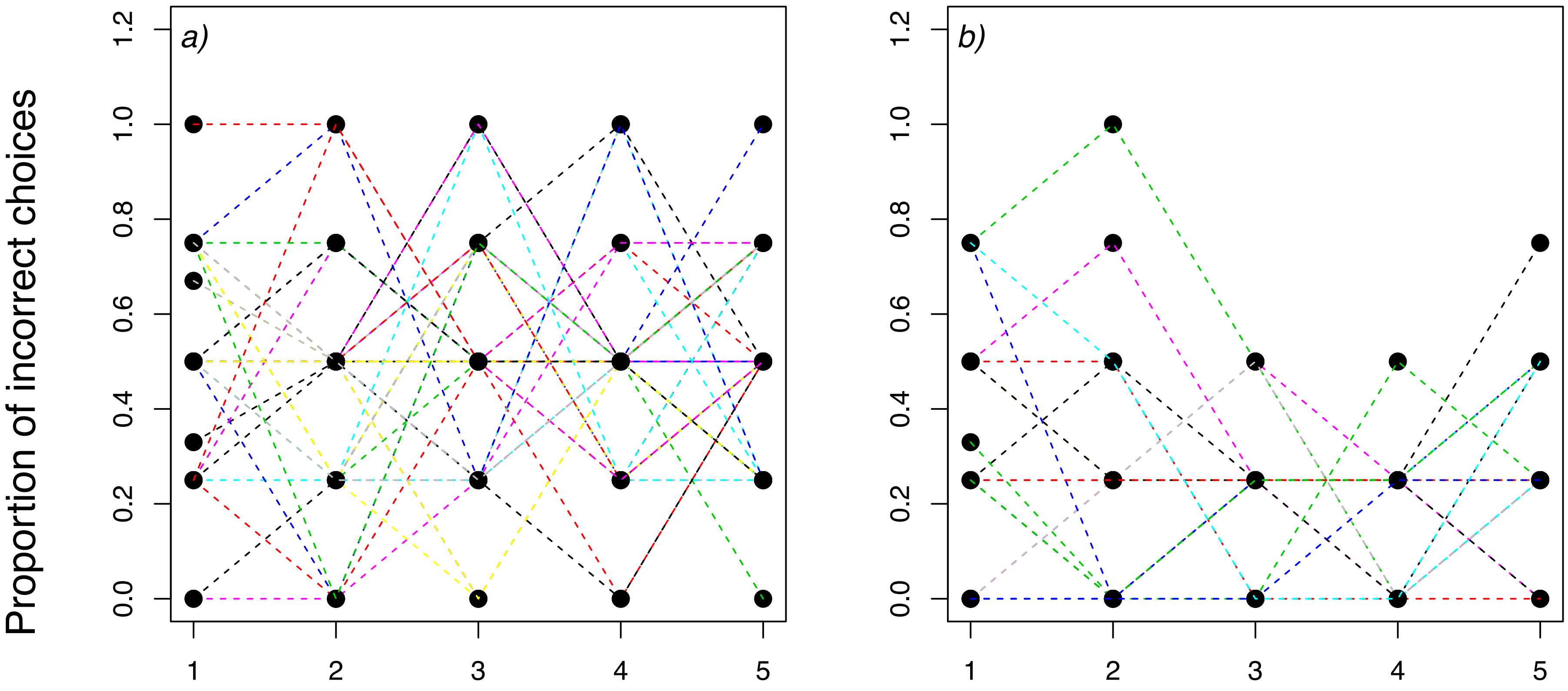


Figure S4. Individual learning curves looking at the mean number of incorrect choices across the 5 learning blocks in (a) non-learner females, (b) non-learner males, (c) learner females, and (d) learner males.

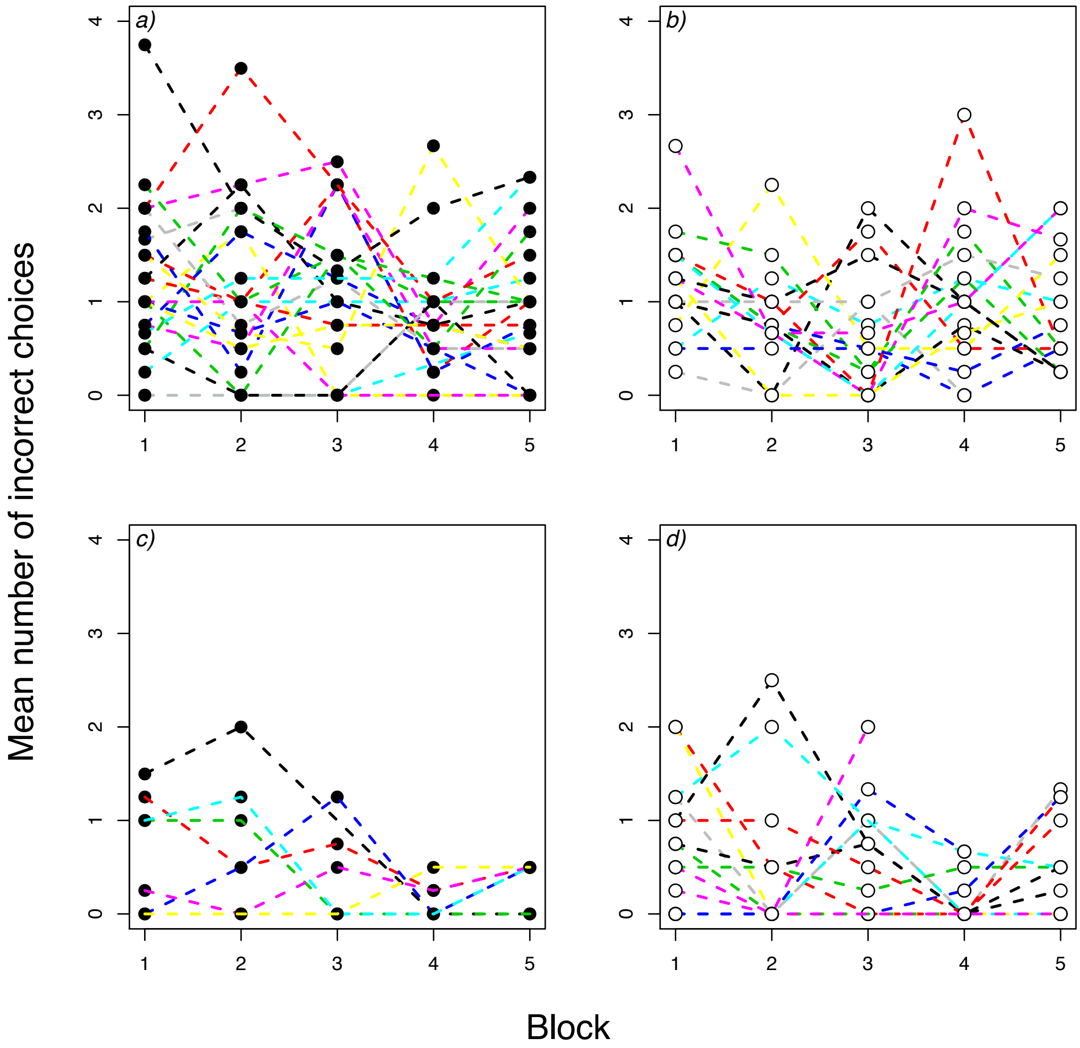


Table S1. Models evaluated testing the effects of sex, body condition (condition) and exploratory (TL and T2ER) and boldness (LATB and NEO) variables on the probability of learning a spatial task. (N) sample size, (K) nº parameters, (*w*) Akaike's weight, (ER) evidence ratio.

Model	**N**	**K**	**AIC_C**	**ΔAIC_C**	**Akaike *w***	**ER**	**Model Formula**
Model 8	56	4	64.89	0	0.399	1	Learning ~ Sex + LATB + $LATB^2$
Model 9	56	5	66.63	1.74	0.167	2.39	Learning ~ Sex + Condition + LATB + $LATB^2$
Model 10	56	5	66.70	1.81	0.161	2.47	Learning ~ Sex + Neo + LATB + $LATB^2$
Model 11	56	5	67.21	2.32	0.125	3.19	Learning ~ Sex + LATB + T2ER + $LATB^2$
Model 12	56	5	67.31	2.42	0.119	3.35	Learning ~ Sex + LATB + L + $LATB^2$
Model 2	56	2	72.00	7.11	0.011	34.99	Learning ~ Sex
Model 3	56	3	73.56	8.67	0.005	76.32	Learning ~ Sex + Condition
Model 1	56	1	73.82	8.93	0.005	86.92	Learning ~ 1
Model 6	56	4	75.33	10.44	0.002	184.93	Learning ~ Sex + Condition + Neo
Model 5	56	4	75.47	10.58	0.002	198.34	Learning ~ Sex + Condition + L
Model 4	56	4	75.52	10.63	0.002	203.36	Learning ~ Sex + Condition + LATB
Model 7	56	4	75.64	10.75	0.002	215.94	Learning ~ Sex + Condition + T2ER

Table S2. Tally of correct/incorrect choices across the 20 trials. We considered lizards to have chosen the correct refuge when they were found already in the safe refuge at the time of simulated predatory attacks or directly run into the safe refuge in response to simulated predatory attacks. The first significant correct/incorrect tallies are underlined for each lizard (with shaded areas reflecting the first consecutive 4 correct choices). Final tallies include all the trials from the first trial in the first significant tally until trial 20. Lizards were categorized as learners (learner, Learning = T; non-learner, Learning = F) when both tallies were significant according to a one-way binomial test (see text for full details).

Lizard	Sex	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20	First tally	Final tally	Learning
43	m	n	n	y	y	y	y	y	y	y	n	y	y	n	y	y	y	n	n	y	n	5/5	13/18	T
95	f	n	n	y	n	n	y	n	y	y	n	y	n	n	y	y	n	y	n	y	n			F
119	f	n	n	y		n	n	n	n	y	y	n	n	y	y	y	y	n	y	n	y			F
172	f	n	y	n	n	y	y	n	y	n	n	n	y	y	n	n	y	y	y	n	y			F
70	m	n	y	y	y	y	n	y	y	n	y	y	y	y	y	n	y	n	y	y	n	11/13	14/19	T
177	m	n	n	n	y	y	y	n	n	y	y	y	y	n	y	n	y	y	y	n	n			F
65	m	n	n	y	n	n	n	n	n	n	n	y	y	y	y	y	y	y	y	n	y	5/5	9/10	T
131	f	y	y	y	n	y	y	n	y	y	y	y	n	n	y	n	y	n	y	y	y			F
139	m		y	y	y	y	y	y	y	y	y	y	n	y	n	y	y	y	y	y	n	5/5	16/19	T
226	f	n	y	y	n	y	y	n	n	y	y	y	y	y	y	y	y	y	y	n	y	5/5	11/12	T
52	m		y	y	y	y	y	y	n	n	y	n	y	y	y	n	y	y	y	y	y	5/5	15/19	T
6	f		y	y	y	y	y	y	y	y	n	y	y	n	y	n	n	n	y	n	n	5/5	12/19	F
207	m		y	n	n	y	y	n	n	y	n	n	y	y	n	n	y	n	n	n	y			F
223	f		y	n	n	y	n	y	n	n	n	y	n	y	y	y	n	n	y	y	y			F
233	f		n	y	y	n	n	y	y	n				n	y	y	n	n	y	n	n			F
105	f	y	y	n	n	n	y	y	y	y	y	n	y	y	n	y	y	y	y	y	n	5/5	12/15	T
215	f		y	y	y	y	n	y	y	y	n	y	n	y	y	y	y	y	y	n	y	10/13	14/19	T
145	m	n	y	y	y	y	y	n	n	y	n	y	y	y	n	y	y	y	y	y	y	5/5	15/19	T
155	m	y	y	y	n	y	y	y	y	n	n	y	y	y	y	n	y	n	n	y	y	9/12	11/16	F
122	m	n	y	n	n	n	y	y	y	n	y	y	n	y	n	y	n	n	n	n	y			F
20	m	n	n	n	y	n	n	y	y	y	n	y	y	n	n	y	y	n	n	n	n			F
208	f	n	n	y	n	n	y	n	n	n	y	y	n	n	n	y	n	y	y	y	n			F
133	f	y	n	n	y	n	y	y	n	y	y	y	y	y	y	y	y	y	y	y	y	5/5	12/12	T
168	m	n	y	y		y	y	y	y	y	y	y	y	n	n	y	y	y	y	n	y	5/5	15/18	T
195	f	n	n	y	n	y	n	y	n	n	y	n	n	y	n	n	y	n	y	n	y			F
36	m	n	n	y	n	y	y	n	y	y	y	y	y	y	n	y	n	y	y	n	n	5/5	9/13	F
193	f	y	n	n	n	y	n	y	n	y	y	n	y	n	n	n	n	y	y	n	n			F
108	f	y	y	y	y	y	y	y	n	y	n	n	n	n	n	n	n	n	n	y	y	5/5	10/20	F

37	m	y	n	n	y	y	n	y	n	n	y	n	y	n	n	y	n	y	y	n	n			F
182	m	y	n	y	n	y	y	y	y	y	y	n	y	y	n	y	y	y	n	y	n	5/5	12/16	T
117	f	y	n	y	n	y	y	n	n	y	n	n	y	y	y	n	n	y	y	y	y			F
219	f	n	y	y	n	y	y	y	y	y	n	n	n	y	n	y	y	n	y	n	y	5/5	10/16	F
179	m	n	n	y	y	y	y	n	n	y	y	y	n	n	n	n	n	y	y	n	y			F
175	f	y	y	n	y	n	y	n	n	n	y	n	y	n	y	n	n	n	n	n	y			F
174	f	y	n	n	n	y	y	n	y	n	y	n	n	n	y	y	y	n	n	y	y			F
28	m	n	y	n	y	n	y	y	y	n	n	y	n	n	y	y	n	y	n	n	y			F
76	m	y	y	y	y	y	y	y	y	y	y	y	y	y	n	y	y	n	y	n	y	5/5	17/20	T
66	m	y	y	n	y	n	y	n	y	n	y	n	y	n	y	y	n	n	n	y	n			F
94	f	n	y	n	y	y	y	n	n	y	n	n	n	y	y	n	y	n	n	y	n			F
46	m	n	y	n	y	n	y	n	n	n	y	y	y	y	y	y	y	y	y	n	y	5/5	10/11	T
16	m	n	n	y	n	y	y	y	y	n	y	n	n	y	y	n	n	y	n	y	y			F
33	m	y	n	n	n	n	n	n	n	y	y	n	y	n	n	n	n	y	y	y	n			F
189	f	y	n	y	n	y	n	y	y	n	n	n	n	y	n	y	y	n	y	y	y			F
99	f	n	y	y	n	y	n	n	y	n	n	n	n	y	n	y	n	n	y	n	y			F
158	m	n	y	n	n	n	y	y	n	y	y	y	y	y	n	n	y	y	n	y	y	5/5	9/12	F
3	f	n	y	y	n	n	y	y	y	y	y	n	y	n	n	y	y	y	n	y	y	5/5	11/15	F
55	m	n	y	n	y	n	y	y	n	y	y	y	n	y	y	y	y	y	n	n	y	5/5	6/8	F
114	f	y	n	y	n	y	y	n	y	y	y	y	n	n	y	y	y	y	y	n	y	9/11	10/13	T
96	f	y	y	y	n	n	n	n	n	n	y	n	y	n	n	y	y	n	n	y	n			F
204	m	n	n	y	y	y	y	y	n	n	y	n	y	y	y	y	y	n	y	y	y	5/5	14/18	T
230	m	y	y	n	n	y	n	y	y	n	y	y	y	y	y	y	y	y	y	n	n	5/5	9/11	T
72	m	n	y	y	y	n	y	y	y	y	y	n	y	y	y	n	y	n	y	y	y	5/5	12/15	T
110	f	n	n	n	y	n	y	n	n	n	y	n	y	y	n	y	n	y	n	n	n			F
73	f	n	y	y	n	n	n	n	y	y	n	y	n	y	y	n	n	y	y	n	n			F
227	f	n	y	y	y	y	y	y	y	y	n	y	y	y	y	n	y	n	n	y	y	5/5	15/19	T
199	m	y	n	y	n	n	y	y	n	n	n	y	y	y	y	n	y	n	n	n	y			F
X	f	y	y	y	y	y	y	y	y	y	y	y	y	n	y	y	y	y	n	y	y	5/5	18/20	T
194	f	n	n	y	y	y	n	n	y	y	y	n	n	y	y	y	n	y	n	y	n			F
41	m	n	y	y	n	y	n	y	n	y	n	y	n	n	n	y	y	y	n	y	y			F
64	m	y	n	n	n	y	n	y	n	y	y	y	y	y	y	y	y	y	n	n	y	5/5	10/12	T
147	m	n	n	n	y	y	y	n	n	y	y	y	n	n	n	y	y	y	n	n	n			F
156	f	y	n	y	n	n	n	n	y	y	n	y	n	y	n	y	n	n	y	y	y			F

Table S3. Models evaluated testing the effects of sex, body condition (condition) and exploratory (TL and T2ER) and boldness (LATB and NEO) variables including two 'borderline' non-learners as learners (M158 and F003, see text above). (N) sample size, (K) nº parameters, (*w*) Akaike's weight, (ER) evidence ratio.

Model	**N**	**K**	**AIC_C**	**ΔAIC_C**	**Akaike *w***	**ER**	**Model Formula**
Model 8	56	4	68.11	0	0.377	1	Learning ~ Sex + LATB + $LATB^2$
Model 10	56	5	69.66	1.55	0.174	2.17	Learning ~ Sex + Neo + LATB + $LATB^2$
Model 9	56	5	69.82	1.71	0.16	2.35	Learning ~ Sex + Condition + LATB + $LATB^2$
Model 11	56	5	70.28	2.17	0.127	2.96	Learning ~ Sex + LATB + T2ER + $LATB^2$
Model 12	56	5	70.38	2.27	0.121	3.11	Learning ~ Sex + LATB + L + $LATB^2$
Model 2	56	2	74.53	6.42	0.015	24.78	Learning ~ Sex
Model 3	56	3	75.89	7.78	0.008	48.91	Learning ~ Sex + Condition
Model 1	56	1	76.17	8.06	0.007	56.26	Learning ~ 1
Model 6	56	4	77.55	9.44	0.003	112.17	Learning ~ Sex + Condition + Neo
Model 4	56	4	78.15	10.04	0.002	151.41	Learning ~ Sex + Condition + LATB
Model 7	56	4	78.16	10.05	0.002	152.17	Learning ~ Sex + Condition + T2ER
Model 5	56	4	78.17	10.06	0.002	152.93	Learning ~ Sex + Condition + L